

DANIEL BARROS VASI WERNER

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Education

University of Michigan - Ann Arbor

Computer Engineering B.S.E.

Expected Graduation Date: Dec. 2025

Coursework

Completed: Advanced Computer Architecture, Embedded Systems Design, Operating Systems, Microarchitecture, Data Structures & Algorithms, Digital Integrated Circuits, Signals & Systems

Ongoing: Advanced Embedded Systems, Real-Time Operating Systems, Digital Signal Processing

Experience

Draper Laboratory

May 2025 – August 2025

Embedded Software Engineer Intern

Cambridge, MA

- Designed platform-agnostic Hardware Abstraction Layer (HAL) in C which implements common application code between STM32, MSP430, and MAX32 platforms.
- Created common HAL interface for all platforms above and STM32 Drivers for GPIO, UART, SPI, I2C, ADC, DAC, Timer, and Flash peripherals.
- Implemented methodology for auto-configuring project code based on platform, board, and specified peripherals using Python auto-coder for C code.
- Developed coding standards, brought up CI/CD pipelines, and ported toolchains from proprietary software to GCC to create internal open-source software for company use.

Projects

Out-of-Order RISC-V Processor (Synthesizable) | SystemVerilog, RISC-V, Verdi

- Designed RV32I processor at RTL with speculative and out-of-order execution, N-way Superscalarness, Early Tag Broadcast, Fast Branch Recovery, Early Branch Resolution, Store Queue (Store → Load forwarding), Non-Blocking L1 D-Cache, GShare Branch Predictor, Branch Target Buffer, and Instruction Prefetching.
- Created rigorous Design Verification testbenches for all modules with additional SystemVerilog assertions.
- Tested processor on a series of RISC-V unit tests and correctly ran programs such as Alexnet and Mergesort.
- Achieved a clock period of 11.5 ns (86.95MHz), and average CPI of 1.3.

Remote Camera Controller | C, STM32, KiCad

- Implemented a control loop that moved a series of pan-and-tilt servos based on joystick inputs and PixyCam location data.
- Created a graphics library with support for loading bitmap images onto an OLED display using I2C communication.
- Made framework for using a wireless remote with XBee to XBee Communication.
- Designed and assembled a PCB and custom-made joystick housing for an ergonomic user-controller.

Error-Correcting Pointers Research | C++, gem5

- Developed Error Correcting algorithm using AN Codes to recover from up to 42 bits of error induced by single bit-flip events.
- Created C++ logic-gate simulation of common hardware modules such as Ripple-Carry Adders, Kogge-Stone Adders, Brent-Kung Adders, and Carry Save Multipliers to test methodology.
- Analyzed drawbacks of proposed hardware solution for Error-Correcting using gem5 simulator to observe 5% program latency impact.

MCU-Controlled Guitar Effects Pedal | C, Arduino, KiCad

- Created a circuit which amplifies incoming guitar signals to be read in by an ADC.
- Built a series of Arduino programs that implemented guitar effects such as distortion, fuzz, and delay.

Ongoing Projects

Electronic Guitar | C, Arduino, Verilog

- Creating an electronic guitar using a series of TFT Touch Displays that map to locations on a fretboard.
- Utilizing an FPGA to do DSP for a FFT visualized on multiple TFT Displays.

MIPS R3000 Multiprocessor | SystemVerilog, Xilinx Vivado

- Creating a modified MIPS R3000 CPU with support for multiprocessors and MSI cache-coherence protocol.
- Implementing a subset of the MIPS I ISA.
- Synthesizing to deploy processor onto a Xilinx Artix-7 FPGA.

Isometric City-Building Strategy Game | C++, OpenGL, SFML

- Creating a game engine with support for custom OpenGL isometric graphics library, audio, player-made assets, and multi-threading.
- Building a real-time simulation of a supply-and-demand economy which interfaces with player-controlled local politics and laws.

Zilog Z80 Microcontroller | C, Z80 Assembly, SystemVerilog, KiCad

- Creating a Microcontroller using Zilog Z80 & FPGA to implement UART and SPI hardware peripherals.

Skills

- **Languages:** C, C++, SystemVerilog, Verilog, ARM Assembly, RISC-V Assembly, Python, Bash, MATLAB, Java
- **Embedded:** FreeRTOS, GPIO, UART, SPI, I2C, ADC, DAC, Timer, PWM, Flash
- **Tools:** KiCad, Git, Xilinx Vivado & Vitis, LTSpice, Synopsys Verdi, Cadence Virtuoso, gem5
- **Misc.:** Electric Guitar/Bass, Logic Pro X, FL Studio, Notion, DaVinci Resolve, Certified Remote Pilot under FAA Part 107, Fluent in English, Portuguese, and Spanish.